



Introduction to Nmap For Beginners

What is Nmap?

Nmap (Network Mapper) is a free, open-source tool for network discovery and security auditing. It scans networks to identify hosts, services, operating systems, and vulnerabilities by sending packets and analyzing responses. Widely used by administrators and security professionals, it supports tasks like inventory, monitoring, and penetration testing. Available on multiple platforms, it offers both command-line and graphical interfaces.

Does Nmap Come Pre-Installed in Kali Linux?

Yes, Nmap is pre-installed by default on Kali Linux, as Kali is a specialized distribution designed for security researchers and penetration testers. It comes with a full suite of Nmap tools, including:

- Nmap: The core command-line network scanning tool.
- Zenmap: The graphical user interface (GUI) for Nmap.
- Ncat: A flexible data transfer and debugging tool.
- Ndiff: A utility for comparing scan results.
- Nping: A tool for packet generation and response analysis.

To confirm Nmap is installed on Kali Linux:

1. Open a terminal.
2. Run the following command:

```
sg-learning@SG: ~  
$ nmap --version  
Nmap version 7.94SVN ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.6 openssl-3.4.1 libssh2-1.11.0 libz-1.3 libpcap-1.10.4 nmap-libdnet-1.12 ipv6  
Compiled without:  
Available nsock engines: epoll poll select
```

If Nmap is not installed (rare for Kali) or you want to ensure you have the latest version, follow the Linux installation steps below.



INSTALLATION GUIDE FOR NMAP

1. INSTALLING NMAP ON LINUX

This section covers installation on Debian/Ubuntu-based distributions (like Ubuntu or Kali) and RPM-based distributions (like Red Hat, CentOS, or Fedora). There are multiple methods to install Nmap: using a package manager, compiling from source, or using RPM packages.

INSTALL NMAP USING PACKAGE MANAGER (DEBIAN/UBUNTU)

This is the easiest and most recommended method for most users.

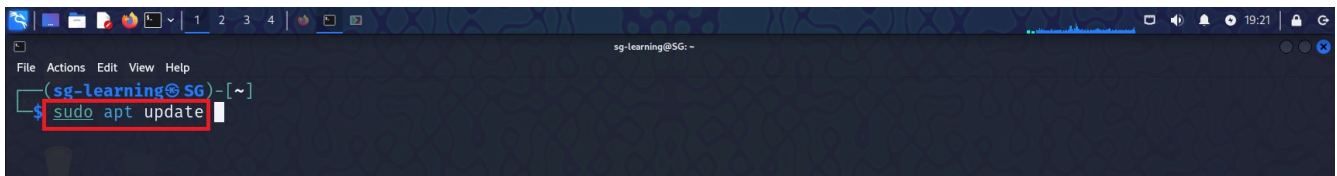
Steps:

1. Open a Terminal:

- On Ubuntu or Kali, open a terminal by pressing Ctrl + Alt + T or searching for "Terminal" in the applications menu.

2. Update Package Lists:

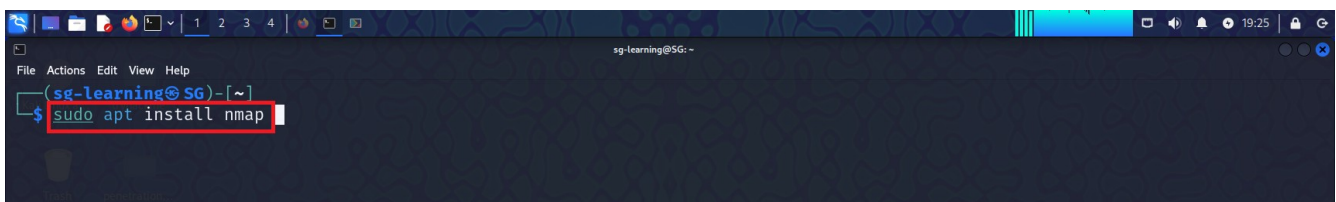
- Ensure your system has the latest package information



```
sg-learning@SG: ~  
$ sudo apt update
```

Enter your password if prompted.

- **Install Nmap:**
- Run the following command to install Nmap and its dependencies



```
sg-learning@SG: ~  
$ sudo apt install nmap
```

Press Y when prompted to confirm the installation.

- **Verify Installation:**



```
sg-learning@SG: ~  
$ nmap --version  
Nmap version 7.94SVN ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.6 openssl-3.4.1 libssh2-1.11.0 libz-1.3 libpcre2-10.42 libpcap-1.10.4 nmap-libdnet-1.12 ipv6  
Compiled without:  
Available nsock engines: epoll poll select
```

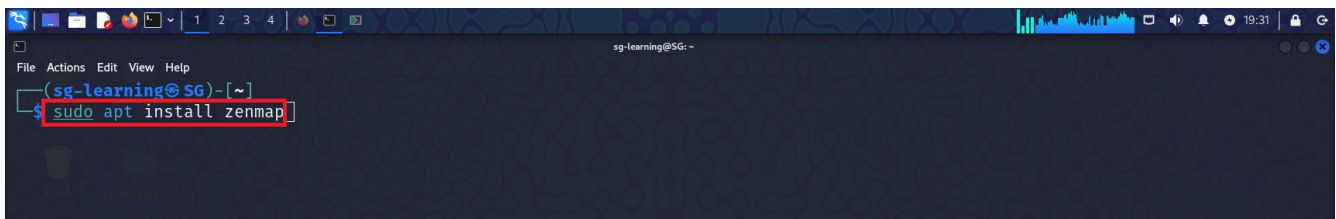


(Optional) Install Zenmap:

-

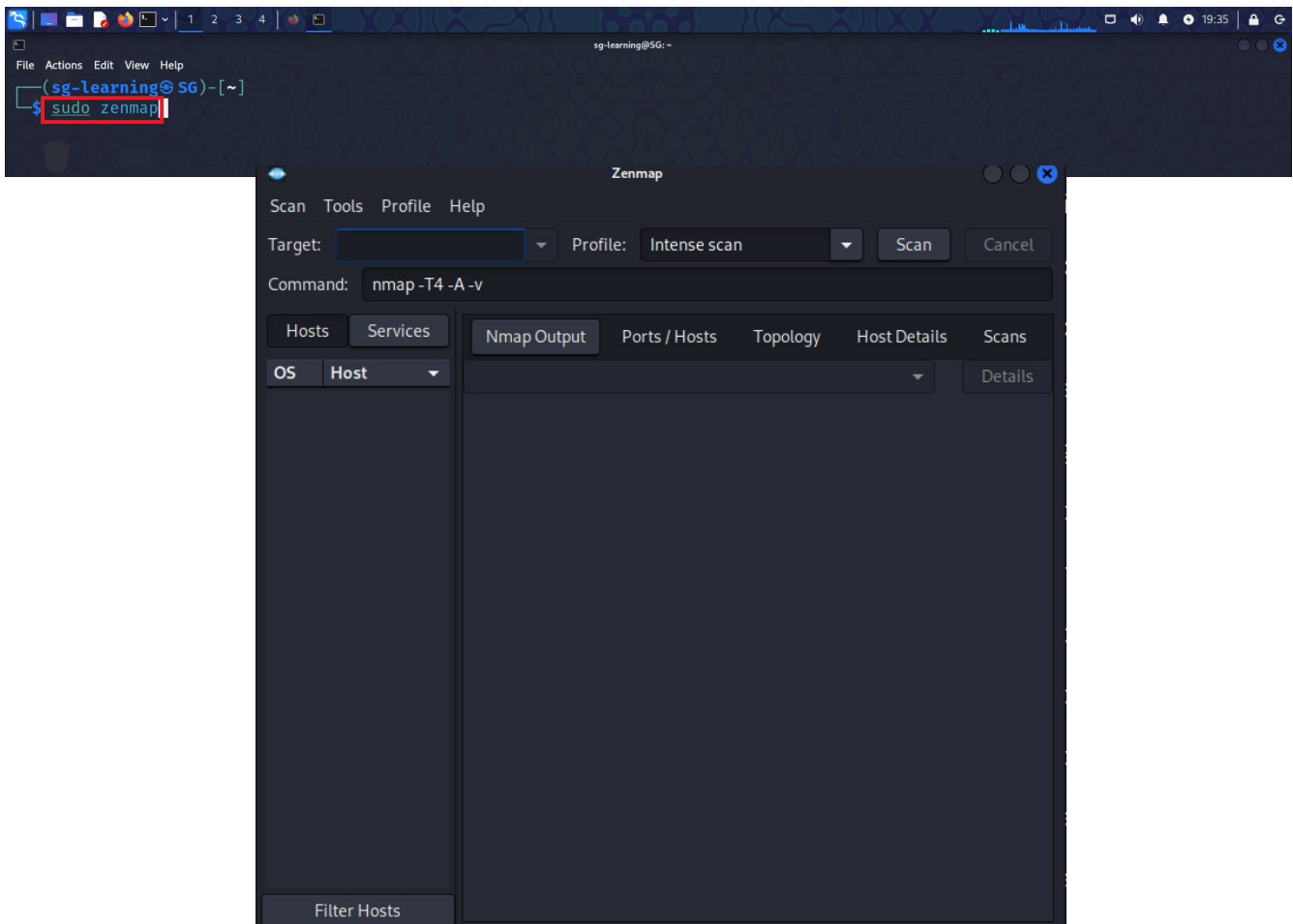
Zenmap is the official Nmap Security Scanner GUI. It is a multi-platform (Linux, **Windows**, **Mac OS X**, **BSD**, etc.) free and open source application which aims to make Nmap easy for beginners to use while providing advanced features for experienced Nmap users. Frequently used scans can be saved as profiles to make them easy to run repeatedly. A command creator allows interactive creation of Nmap command lines. Scan results can be saved and viewed later. Saved scan results can be compared with one another to see how they differ. The results of recent scans are stored in a searchable database.

Install it separately if desired:



```
sg-learning@SG: ~  
$ sudo apt install zenmap
```

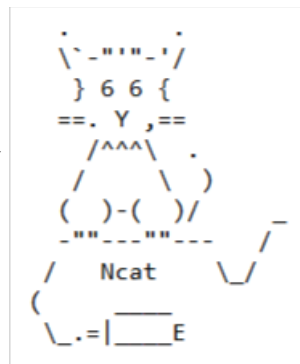
Launch Zenmap by typing zenmap in the terminal or finding it in the applications menu.



Troubleshooting:

- If you get a "package not found" error, ensure your package repositories are correctly configured.
- Check `/etc/apt/sources.list` or run `sudo apt update` again.
- If Nmap is outdated, consider compiling from source (Method 3) or adding the Nmap repository for the latest version.

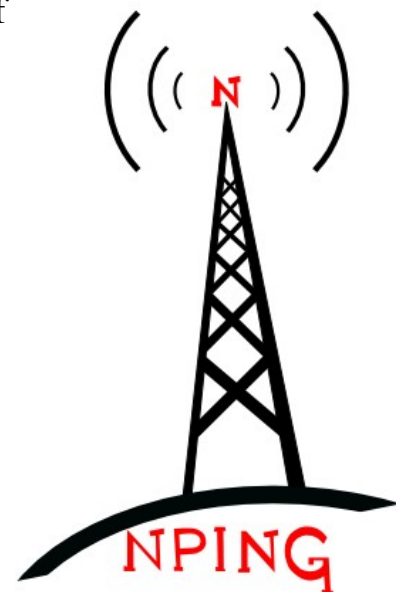
Ncat is a feature-packed networking utility which reads and writes data across networks from the command line. Ncat was written for the Nmap Project as a much-improved reimplement of the venerable [Netcat](#). It uses both TCP and UDP for communication and is designed to be a reliable back-end tool to instantly provide network connectivity to other applications and users. Ncat will not only work with IPv4 and IPv6 but provides the user with a virtually limitless number of potential uses.



Among Ncat's vast number of features there is the ability to chain Ncats together, redirect both TCP and UDP ports to other sites, SSL support, and proxy connections via SOCKS4 or HTTP (CONNECT method) proxies (with optional proxy authentication as well). Some general principles apply to most applications and thus give you the capability of instantly adding networking support to software that would normally never support it

Ndiff is a tool to aid in the comparison of Nmap scans. Specifically, it takes two Nmap XML output files and prints the differences between them: hosts coming up and down, ports becoming open or closed, and things like that. Ndiff can produce output in human-readable text or machine-readable XML formats. Many people like to scan their networks regularly (daily, weekly, etc.) and then use ndiff to easily detect any changes. The scans, ndiff run, and emailed report are often automated using tools such as cron on UNIX or the Scheduled Tasks tool on Windows. Ndiff is also used by the [Zenmap GUI](#) to compare scan results.

Nping is an open source tool for network packet generation, response analysis and response time measurement. Nping can generate network packets for a





wide range of protocols, allowing users full control over protocol headers. While Nping can be used as a simple ping utility to detect active hosts, it can also be used as a raw packet generator for network stack stress testing, ARP poisoning, Denial of Service attacks, route tracing, etc. Nping's novel [echo mode](#) lets users see how packets change in transit between the source and destination hosts. That's a great way to understand firewall rules, detect packet corruption, and more.

BASIC NMAP SYNTAX

The general structure of an Nmap command is:

`nmap [Scan Type(s)] [Options] [Target(s)]`

[Scan Type(s)]: Specifies the type of scan (e.g., TCP SYN, UDP, ping).

[Options]: Modifies scan behavior (e.g., port range, output format).

[Target(s)]: The host(s) or network(s) to scan (e.g., IP address, hostname, CIDR range).

Basic Scan (Default TCP Scan):

```
nmap 192.168.1.1
```

Explanation:

- Scan 192.168.1.1 for open TCP ports.
- No scan type specified, so Nmap defaults to **-sS** (SYN scan) if run with root privileges, or **-sT** (TCP connect scan) otherwise.

Host discovery:

Who is online?

Before to answer this question let we understand (IP range using -i: if you want to scan all the IP address from 192.168.0.1 to 192.168.0.1-10, you can write in this way 192.168.0.1-10).

IP subnet, you can express it as 192.168.0.1/12, and this would be equivalent to 192.168.0.0-255

Hostname (you can also specify your target by hostname, for example google.com).

To discover the online hosts on a network. Nmap offers the **-sn** option (ping scan). However, don't expect this to be limited like ping. Let's see this in real live practice (before we get started we are either running nmap as root or using sudo because we don't want to restrict Nmap's abilities with our account privileges. Running nmap as a local (non-root) user would limit us to fundamental type of scan such as ICMP echo and TCP connect scans; we will revisit this at the end).

Scanning a "Local" Network

```
nmap -sn 192.168.11.0/24
```

The command `nmap -sn 192.168.11.0/24` performs a host discovery scan (also called a "ping scan") using Nmap. Below is a concise explanation of what it does, why it's used, and how it works.



COMMAND BREAKDOWN

- **nmap**: The Nmap command-line tool for network scanning.
- **-sn**: Disables port scanning and performs only **host discovery**. It checks which hosts are online (up) without scanning their ports.
- **192.168.11.0/24**: Specifies the target network, a subnet with 256 IP addresses (from 192.168.11.0 to 192.168.11.255).

WHAT IT DOES

- Sends probes (e.g., ICMP echo requests, TCP SYN/ACK packets) to determine which hosts in the 192.168.11.0/24 subnet are **online** (responding).

PORT SCAN WITH NMAP

A port scan checks which ports on a target are **open**, **closed**, or **filtered** to identify running services (e.g., web server on port 80).

```
sudo nmap -sS -p 80,443 192.168.11.1
```

Explanation:

- **-sS**: TCP SYN scan (stealth, requires sudo).
- **-p 80,443**: Scans ports 80 (HTTP) and 443 (HTTPS).
- **192.168.11.1**: Target IP.

CHECKING LISTENING PORTS

"Listening" means a service is actively accepting connections on a port. Nmap identifies listening ports as **open** in the scan results.

HOW TO CONFIRM LISTENING PORTS

1. Run a Port Scan:

- Use the above command or scan more ports:

```
sudo nmap -sS -p 1-1000 192.168.11.1
```

Scans ports 1–1000 to find open (listening) ports.

Interpret Results:

- **Open**: A service is listening (e.g., web server on port 80).
- **Closed**: No service is listening.
- **Filtered**: A firewall may be blocking the port, so Nmap can't determine the state.

**(Optional) Service Detection:**

- Add -sV to identify services listening on open ports

```
sudo nmap -sS -sV -p 80,443 192.168.11.1
```

NOTES

- **Privileges:** Use sudo for -sS scans (Linux) or run as admin (Windows).
- **Legal:** **Only scan targets you own or have permission to scan.**
- **Common Ports:** 80 (HTTP), 443 (HTTPS), 22 (SSH), 3389 (RDP).
- **Filtered Ports:** If many ports are filtered, try -Pn to skip host discovery

Topic	Example Command	Purpose
Scan a single IP	<code>nmap 192.168.1.1</code>	Scan a specific host
Scan multiple IPs	<code>nmap 192.168.1.1 192.168.1.2</code>	Scan multiple targets
Scan a subnet	<code>nmap 192.168.1.0/24</code>	Discover all devices in LAN
Disable DNS resolution	<code>nmap -n 192.168.1.1</code>	Faster scanning
Scan specific ports	<code>nmap -p 22,80,443 192.168.1.1</code>	Only scan selected ports

Service & Version Detection (Intermediate)**How to Use Nmap to Check Ports**

Nmap is a versatile command-line tool that performs powerful **port** scans. To conduct a simple scan, use the **nmap command** without any options:

```
nmap [target]
```

The target is either a domain name or an IP address. For example, to scan the website `scanme.nmap.org`, use:



```
sg-learning@SG: ~  
File Actions Edit View Help  
(sg-learning@SG)-[~]  
$ nmap scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-07 20:51 EAT  
Nmap scan report for scanme.nmap.org ( )  
Host is up (0.20s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 993 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
2000/tcp  open  cisco-sccp  
9929/tcp  open  nping-echo  
31337/tcp open  Elite  
  
Nmap done: 1 IP address (1 host up) scanned in 6.45 seconds
```

The command without any options scans the most common 1000 ports. Nmap can scan a single port, a port range, or all ports on a target. Below are step-by-step instructions on how to use Nmap to scan for open ports in various ways.

Scan a Single Port

To use Nmap to scan a single port on a target, use the following syntax: `nmap -p [port] [target]`

For example:

```
(sg-learning@SG)-[~]  
$ nmap -p- scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-07 21:00 EAT
```

The scan takes time to complete. The command performs a comprehensive scan of all port numbers. It thoroughly assesses the target network and shows open ports for the provided location.

Alternatively, to scan all standard ports, use the fast scan with the `-F` tag:



```
sg-learning@SG: ~  
File Actions Edit View Help  
(sg-learning@SG)-[~]  
$ nmap -F scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-07 21:03 EAT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.21s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 96 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 3.43 seconds
```

The fast scan checks the 100 most common ports. This method reduces the scan time, which is helpful with large networks.

Scan a Series of Ports

The Nmap tool offers several different ways to scan multiple ports. The examples below demonstrate how to use the tool.

To scan a port range, use the `-p` option, the starting and ending port numbers:

For example, to scan the first 200 ports, use:

```
nmap -p 1-200 scanme.nmap.org
```



```
sg-learning@SG: ~  
File Actions Edit View Help  
(sg-learning@SG)-[~]  
$ nmap -p 1-200 scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-07 21:08 EAT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.21s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 196 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 2.85 seconds
```

To scan multiple specific ports, use a comma-separated list:

```
nmap -p 22,53,80 scanme.nmap.org
```

```
sg-learning@SG: ~  
File Actions Edit View Help  
(sg-learning@SG)-[~]  
$ nmap -p 22,53,80 scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-07 21:11 EAT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.18s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.55 seconds
```

Combine the two methods to scan both specific port numbers and ranges. For example:

```
nmap -p 22,80,100-200 scanme.nmap.org
```



```
sg-learning@SG: ~  
File Actions Edit View Help  
(sg-learning@SG)-[~]  
$ nmap -p 22,80,100-200 scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-07 21:14 EAT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 101 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

Scanning targeted ports or a port range shortens the scan time.

Common Ports

There are many standardized ports associated with specific services. Use the list below as a reference for these ports and their related service:

- **21 (FTP)**. File transfer protocol.
- **22 (SSH)**. Secure shell.
- **25 (SMTP)**. Simple mail transfer protocol.
- **53 (DNS)**. Domain name system.
- **67, 68 (DHCP)**. Dynamic host configuration protocol.
- **80 (HTTP)**. Hypertext transfer protocol.
- **110 (POP3)**. Post office protocol version 3.
- **123 (NTP)**. Network time protocol.
- **143 (IMAP)**. Internet access message protocol.
- **443 (HTTPS)**. Hypertext transfer protocol secure.
- **465 (SMTPS)**. SMTP secure.
- **631 (CUPS)**. Common **Unix** printing system.
- **993 (IMAPS)**. **IMAP** secure.
- **995 (POP3S)**. **POP3** secure.
- **3306 (MySQL)**. MySQL database server.
- **3389 (RDP)**. **Remote desktop** protocol.
- **8080 (HTTP alternate)**. HTTP alternate, used for **proxy servers**.



NMAP

SG-Learning